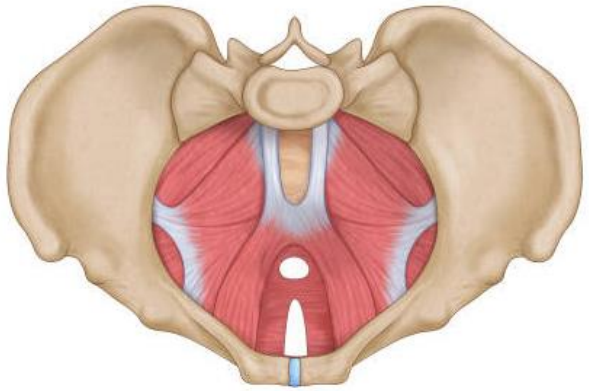
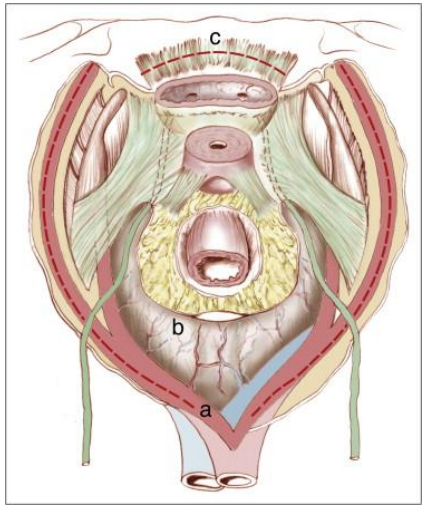


Pelvis and perineum



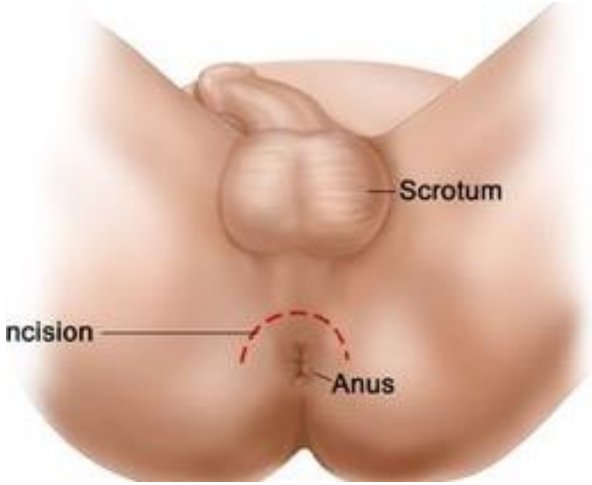
BY:
Dr. NALI Abdulkader

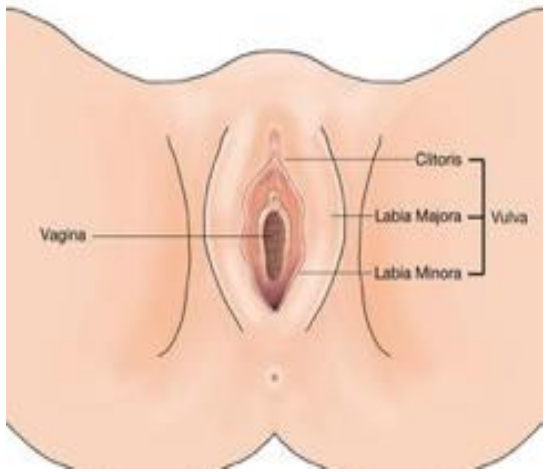


Outlines

- Components of the Bony Pelvis and sub- division of the pelvis (Greater and lesser pelvis)
- Structures forming the walls of the pelvis
- Types of female bony pelvises
- Pelvic floor under the following headings. A. location B. Parts. 3. Gaps
- Define the : A. perineal body b. Anococcygeal body
- Boundaries and contents of the Ischiorectal fossa and anal triangle

Pelvis

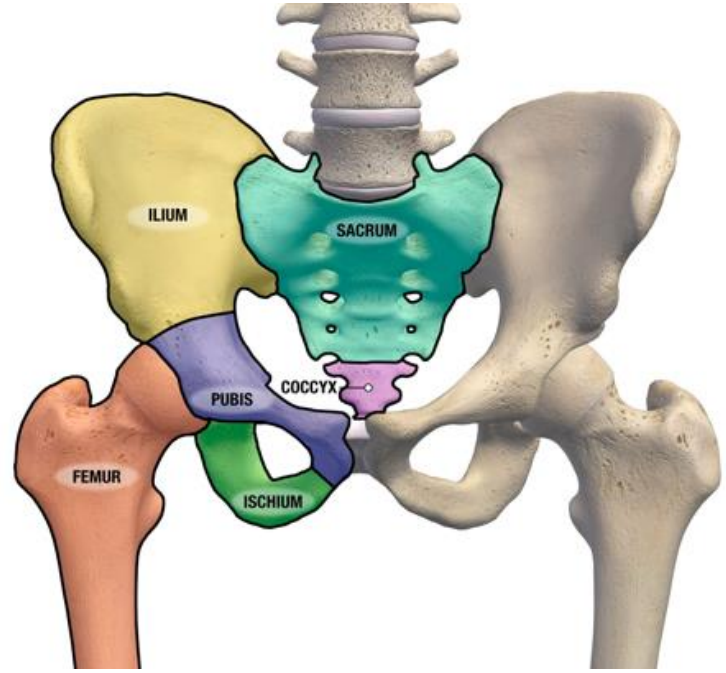




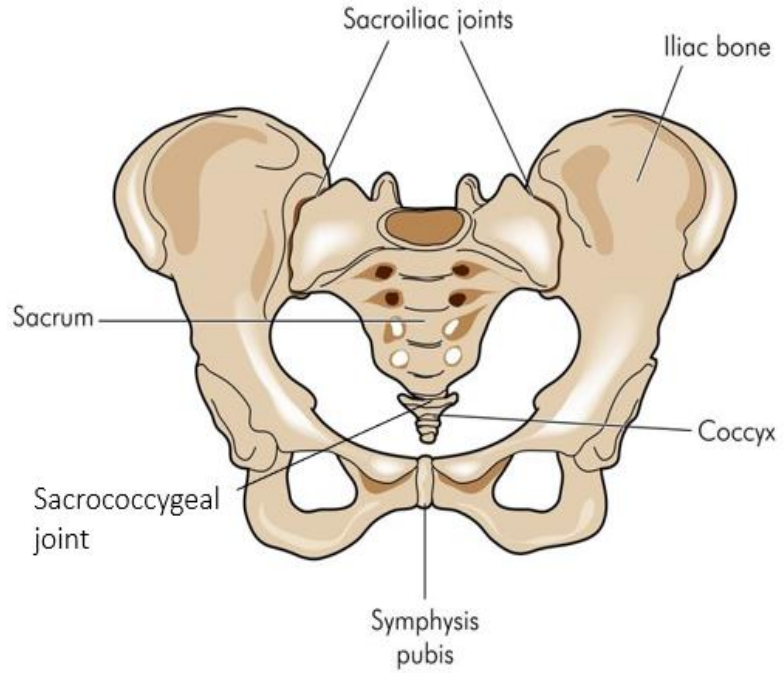


- **The pelvis** is the lower part of the trunk located between the abdomen and the lower limb (has the bony pelvis and cavity)
- **The perineum** is an **anatomical** region in the **pelvis**. It is located between the thighs, contains external genitalia

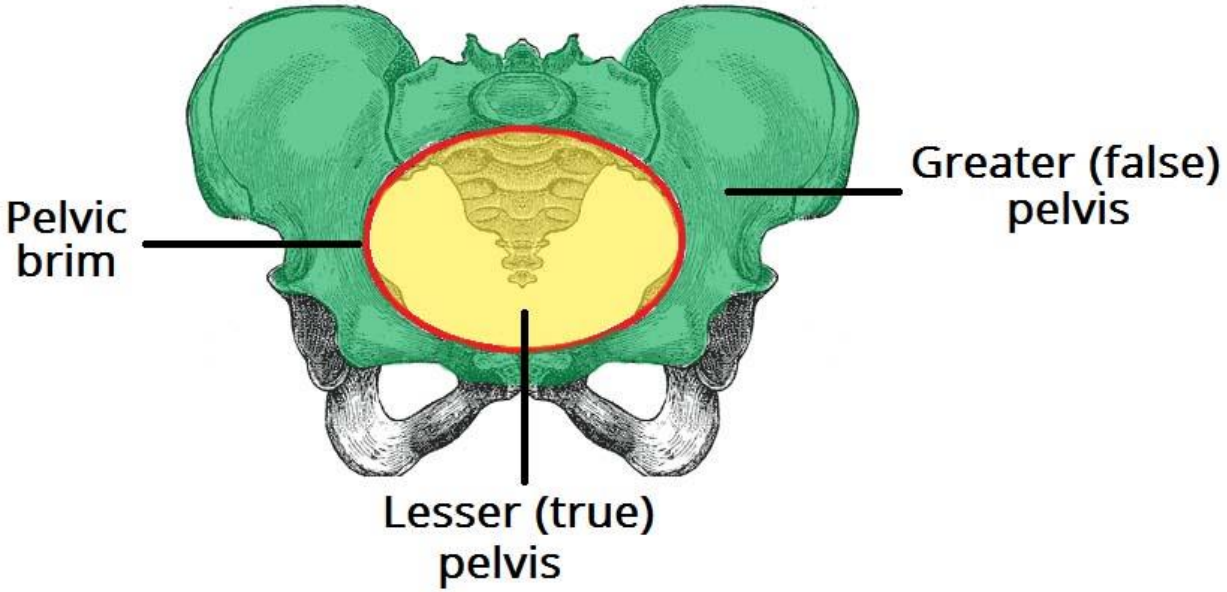
Bony pelvis



- Bones of the pelvis composed four bones two hip bone, sacrum and coccyx
- It provides a strong and stable connection between the trunk and lower limb
- The two hip bone articulates **anteriorly** at the symphysis pubis and **posteriorly** with the sacrum at sacroiliac joint



Joints of the pelvis : SYMPHYSIS PUBIS, ACROILIAC JOINTS and SACROCOCCYGEAL JOINT



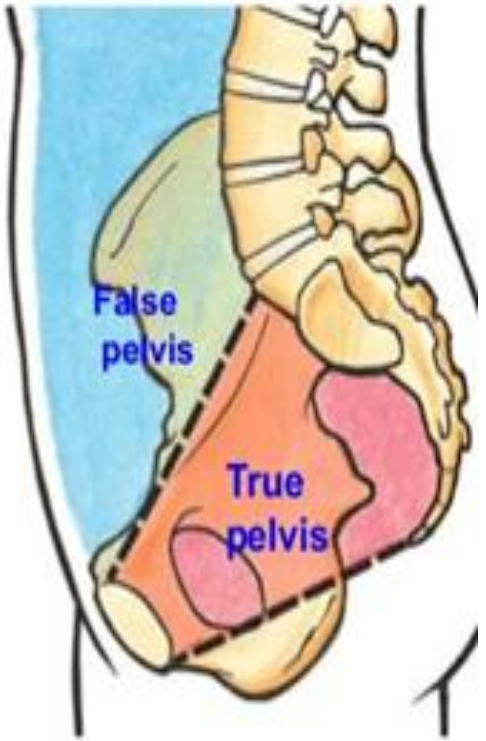
- Pelvic inlet (= pelvic brim)
  - also called superior pelvic aperture
- Pelvic outlet
  - also called inferior pelvic aperture
  - closed by the pelvic diaphragm

CLASSIFICATION OF PELVIS

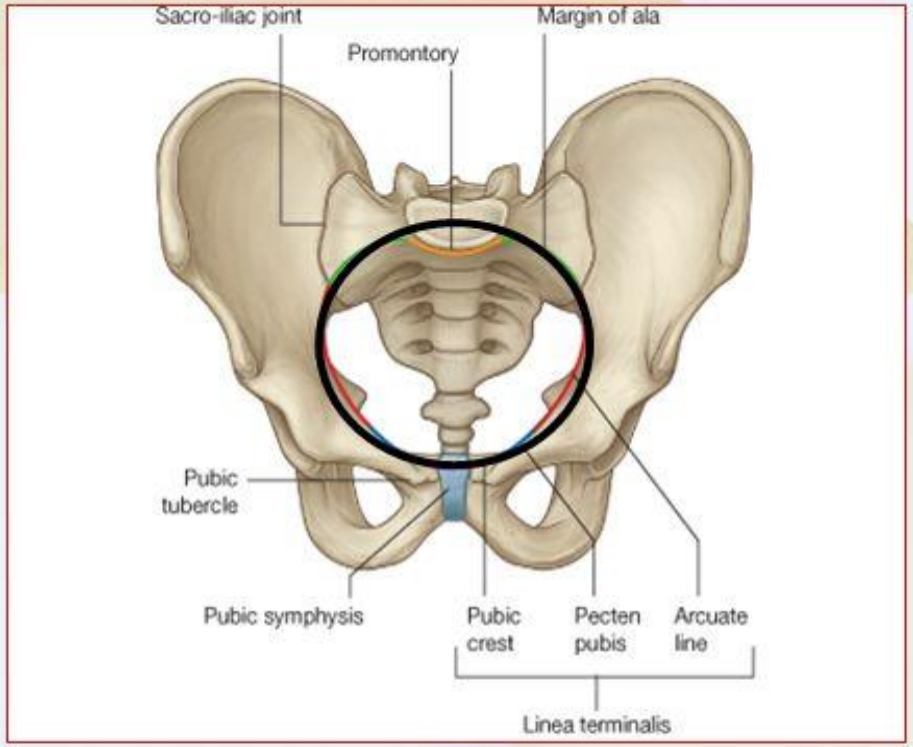
Divided into:

1) False pelvis (pelvis major; greater pelvis)
   - Part of abdominal cavity

2) True pelvis (pelvis minor; lesser pelvis)
   - Is the true pelvic cavity
   - Lies inferior to pelvic brim/pelvic inlet



PELVIC INLET

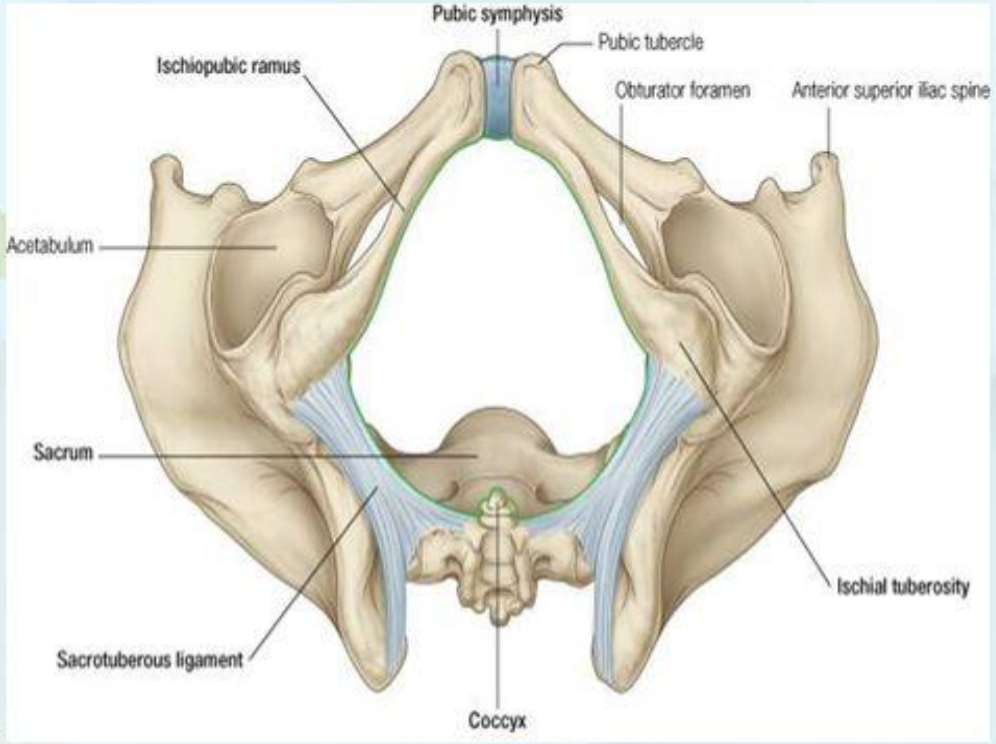


Anteriorly: Symphysis pubis.

Posteriorly: Promontory of sacrum, ala of sacrum.

Laterally: Ileopectineal (arcuate) lines.

PELVIC OUTLET



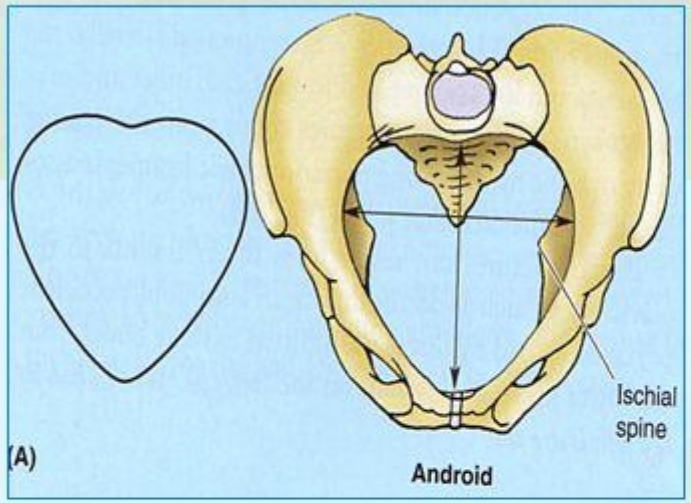
Anteriorly: Symphysis pubis.

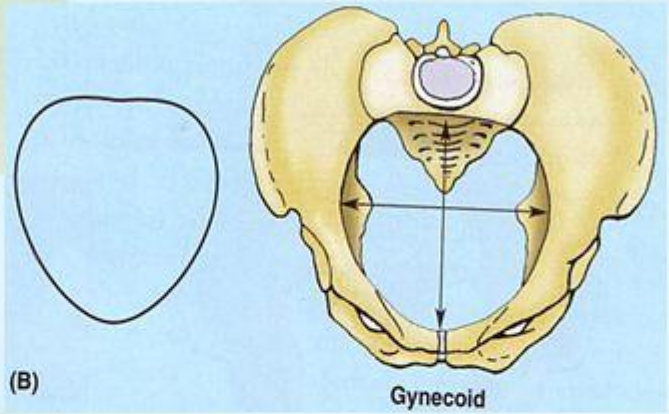
Posteriorly : **Coccyx,**

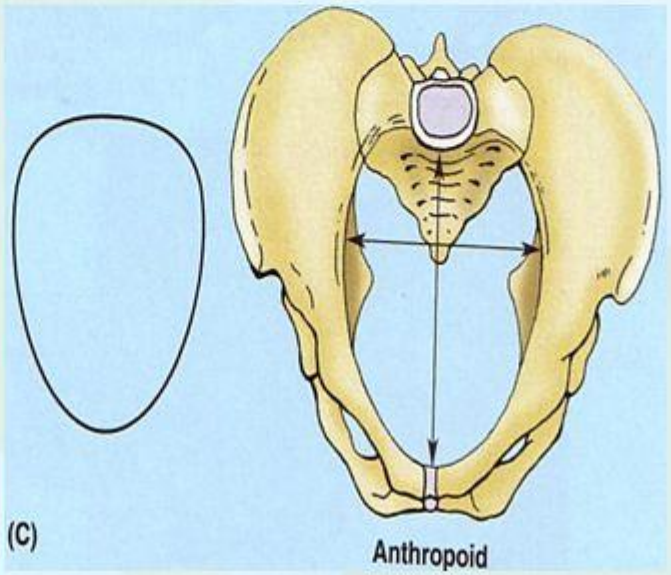
Anterolaterally: **ischiopubic ramus**

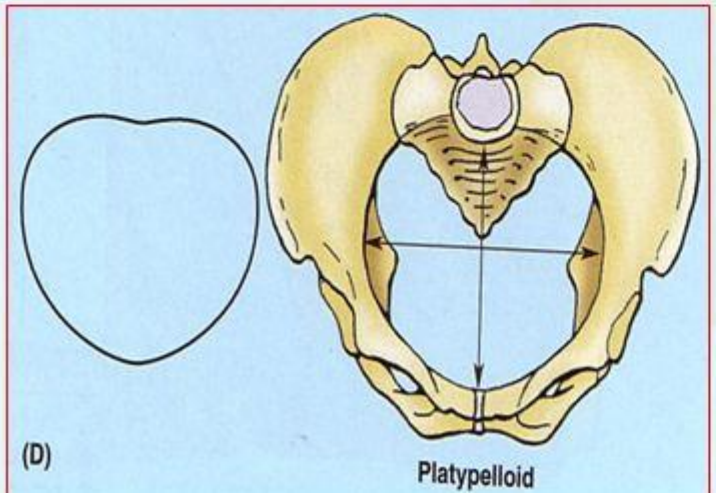
Posterolaterally: Sacrotuberous ligament

Types of Female Bony Pelvis









Information of the shape and dimensions of the female pelvis is of **great importance for obstetrics,** because it is the bony canal through which the child passes during birth.

Features	Gynecoid	Android	Anthropoid	Platypelloid
Pelvic inlet shape	Rounded	Heart-shaped	Oval (anteroposterior)	Wide, flattened oval
Pelvic cavity	Roomy	Funnel-shaped	Deep and narrow	Shallow
Subpubic angle	Wide	Narrow	Moderately wide	Wide
Sacrum	Broad, well curved	Inclined forward	Long and narrow	Short and broad
Obstetric significance	Most favorable for vaginal delivery	Increased risk of difficult labor	Vaginal delivery often possible	Engagement of fetal hea may be difficult

Important anatomical facts:

- **Gynecoid:** Classical female pelvic type, most favorable for childbirth.
- **Android:** Resembles the typical male pelvis.
- **Anthropoid:** Anteroposterior diameter exceeds transverse diameter.
- **Platypelloid:** Transverse diameter exceeds anteroposterior diameter.

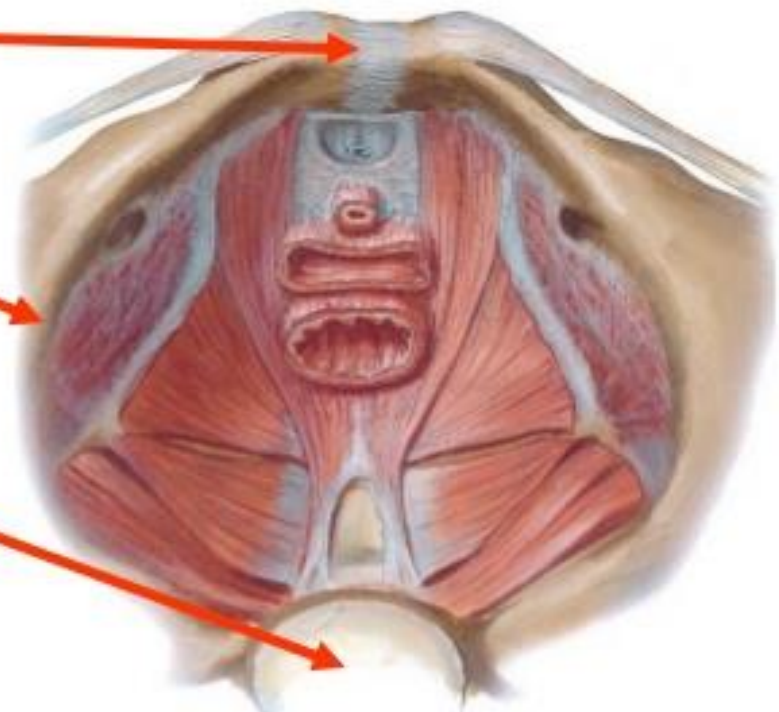
WALL OF PELVIC CAVITY

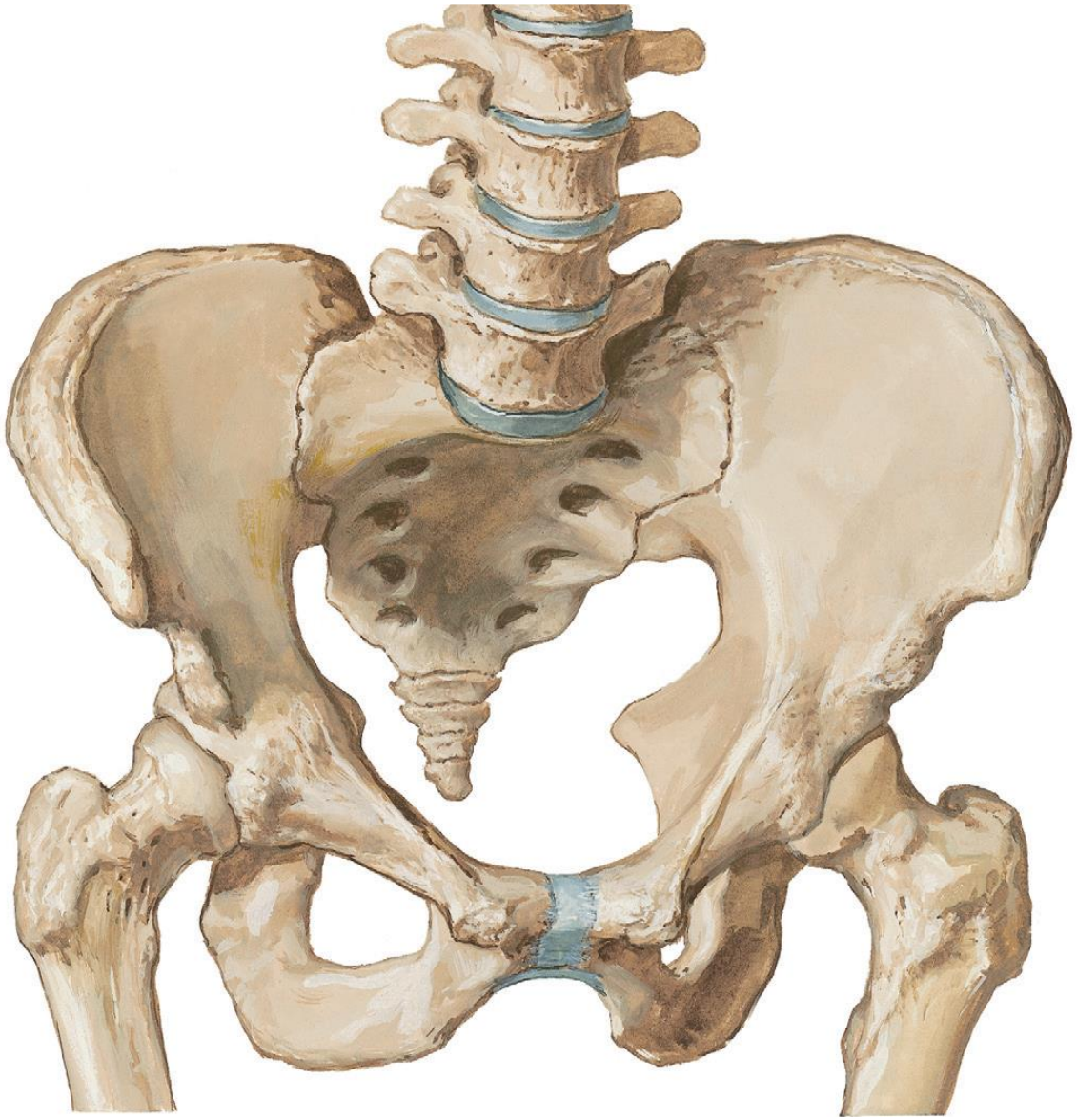
1) Anterior pelvic wall

2) Lateral pelvic wall

3) Posterior wall

4) Pelvic floor





Anterior pelvic wall

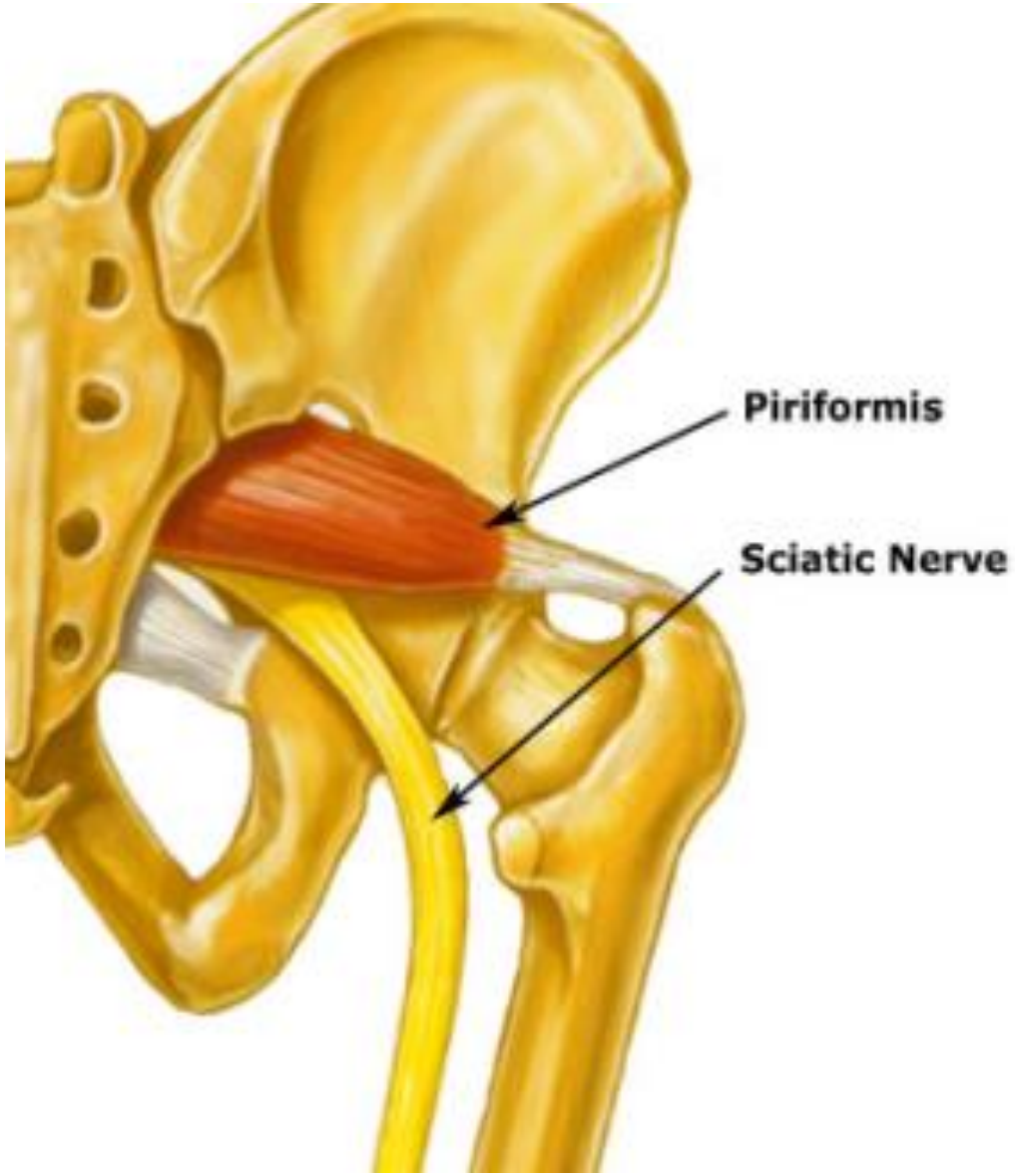
- The wall is shallowest wall formed by
  - ***Symphysis pubis*** which is secondary cartilaginous joint (fibrocartilaginous joint)

    The joint is reforced by two ligaments (sup and inf)
  - ***Pubic rami***

The posterior wall :

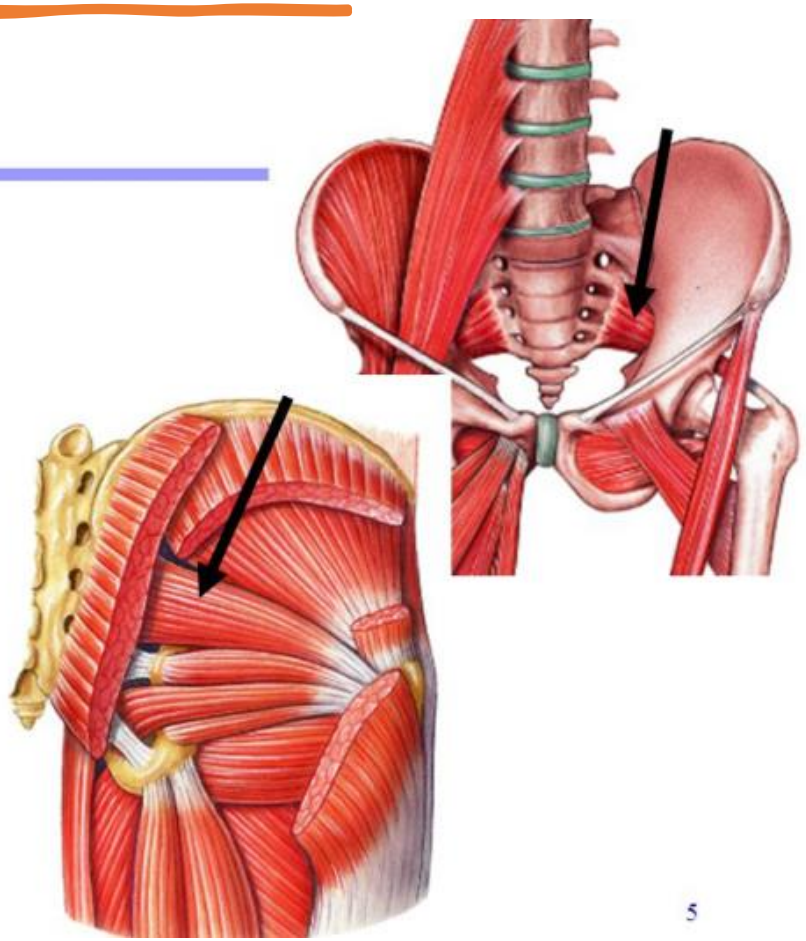
is formed by : ***Sacrum & coccyx 2- Piriform muscle***

3- fascia

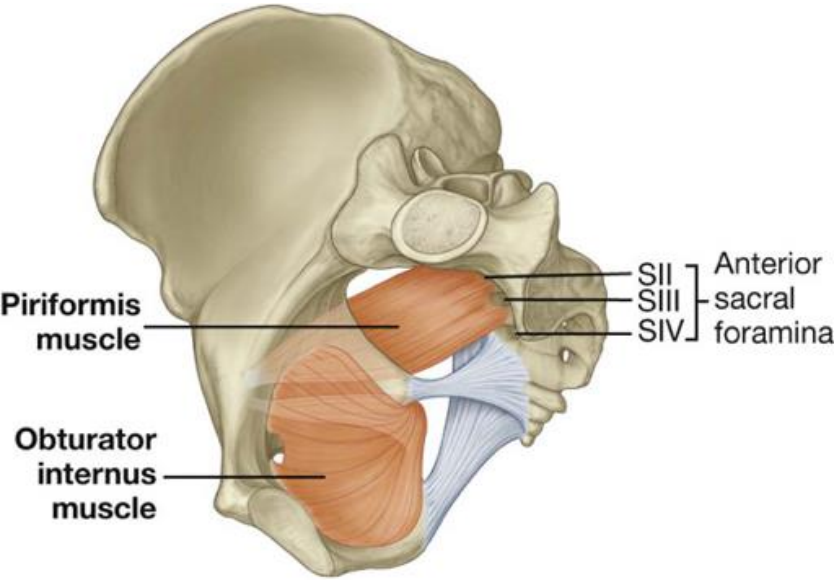


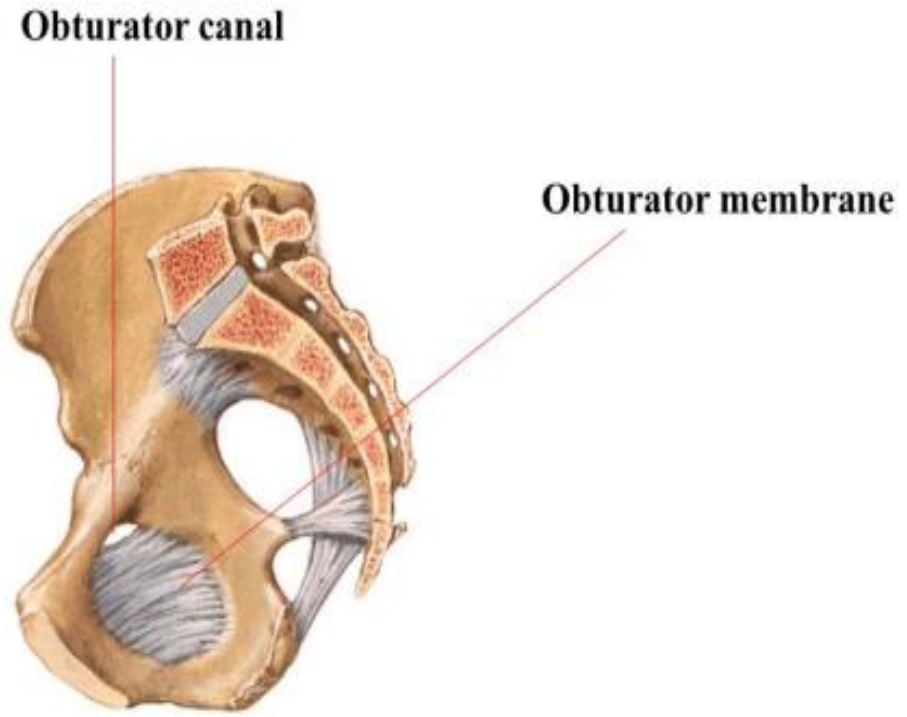
Piriformis

- Origin: Pelvic surface of sacrum
- Insertion: Greater trochanter of femur
- Action: Rotates femur laterally



- **The lateral wall:** is formed by
- ***1- Hip bone 2- Obturator m.m***
- ***3- Obturator muscle and facia***
- ***4- Sacro-tuberous and Sacro- spinous lig***





Obturator Internus

Origin: Inner surface of **obturator membrane**

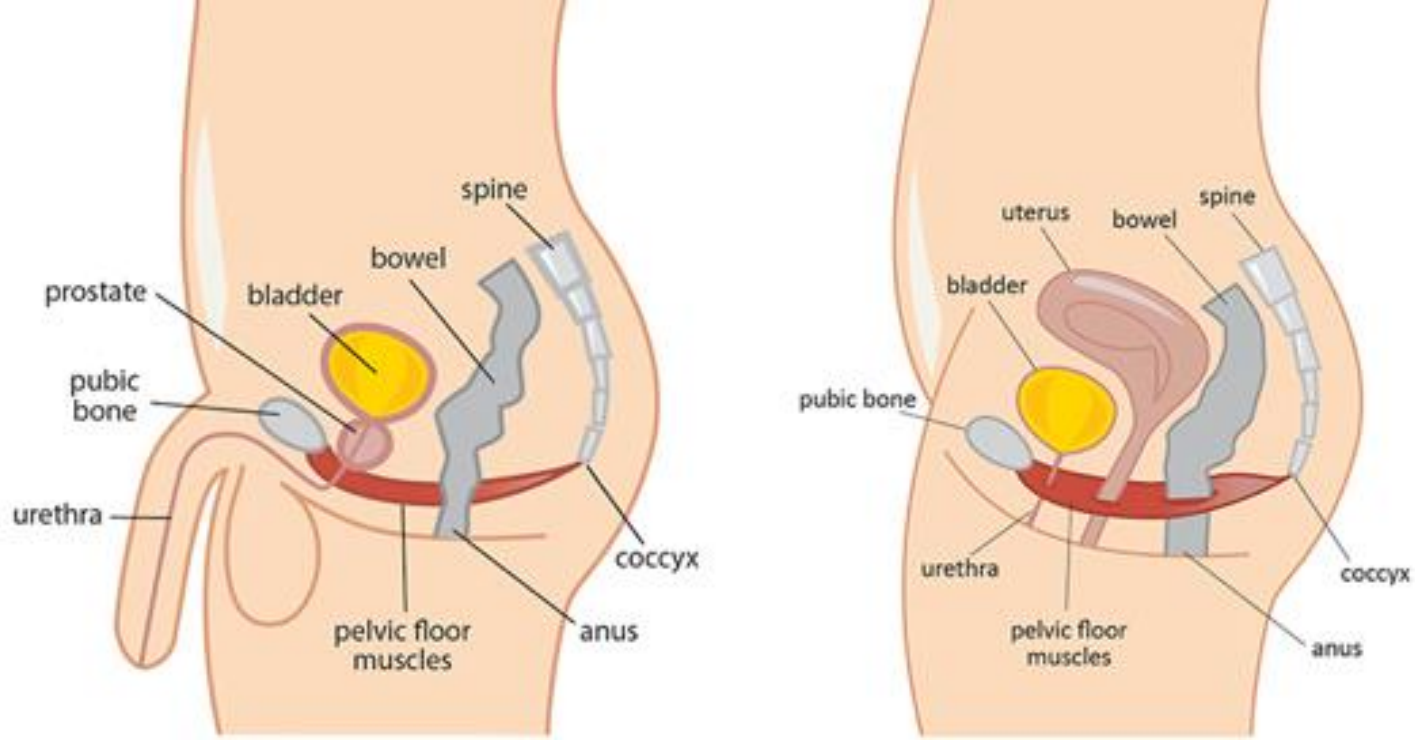
Insertion: **medial border** of greater trochanter

Nerve supply: nerve to obturator internus (L4,S1)

Action:

Lateral rotator of thigh

Pelvic floor



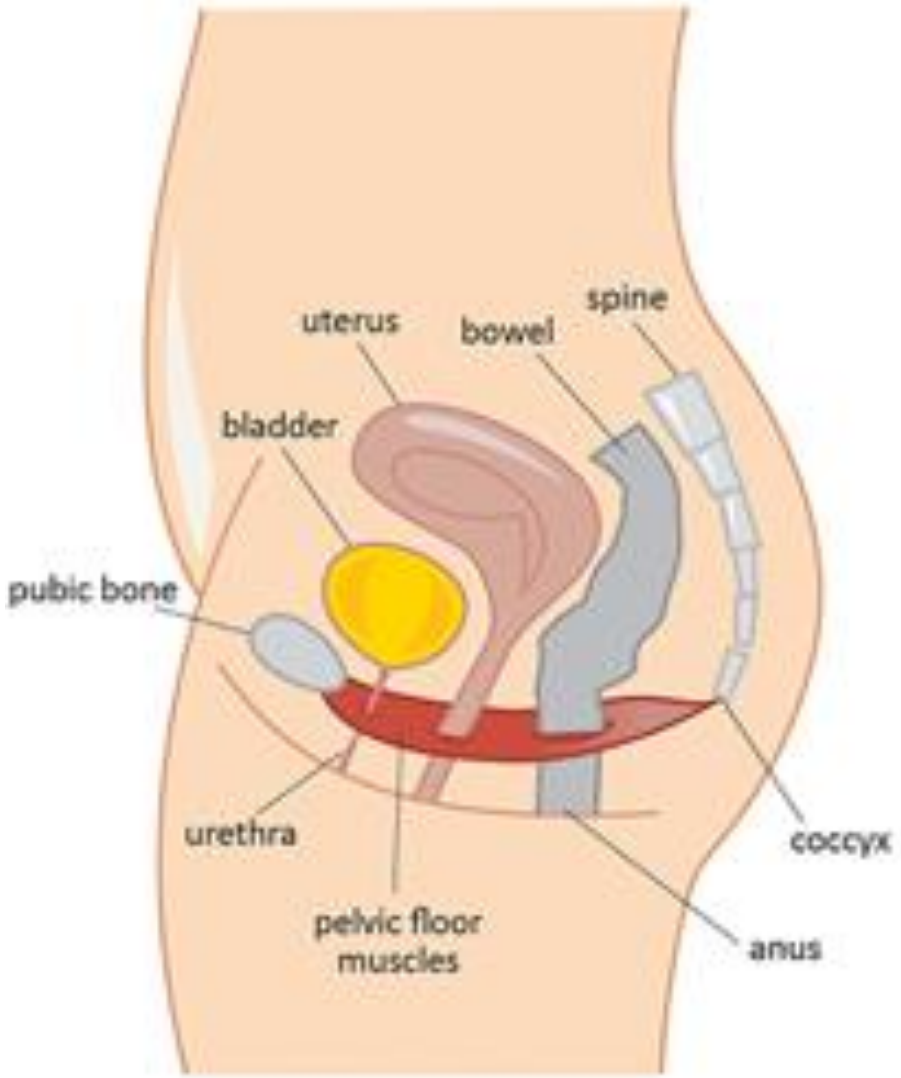
- The pelvic floor is a funnel-shaped structure, separating the pelvic cavity from the perineum inferiorly

❖ Gaps of the Pelvic floor

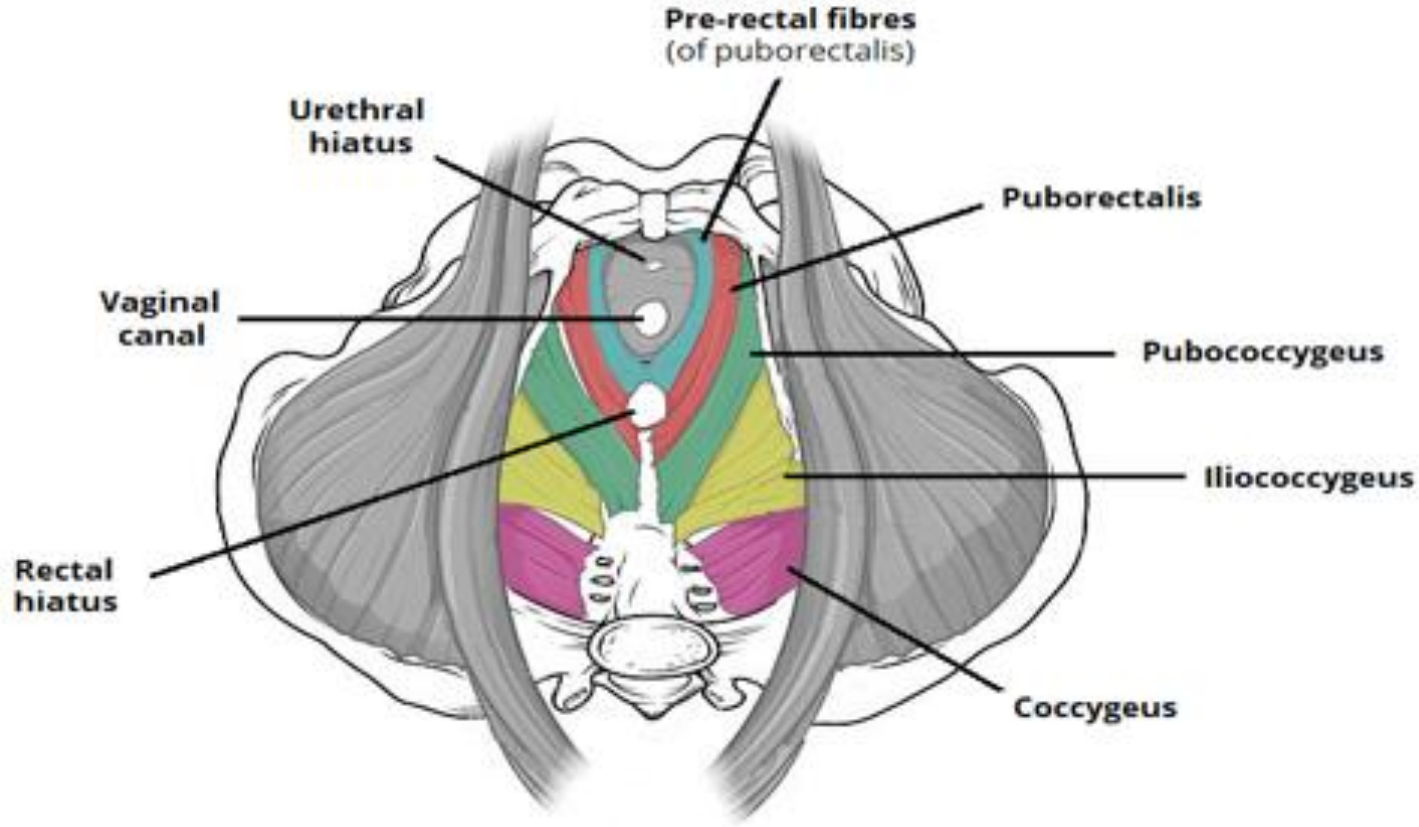
- Urogenital hiatus – an anteriorly situated gap, which allows passage of the urethra (and the vagina in females).
- Rectal hiatus – a centrally positioned gap, which allows passage of the anal canal.

Pelvic diaphragm

- There are three main components of the pelvic floor:
  - Levator ani muscles (largest component).
  - Coccygeus muscle.
  - Fascia coverings of the muscles.



The levator ani

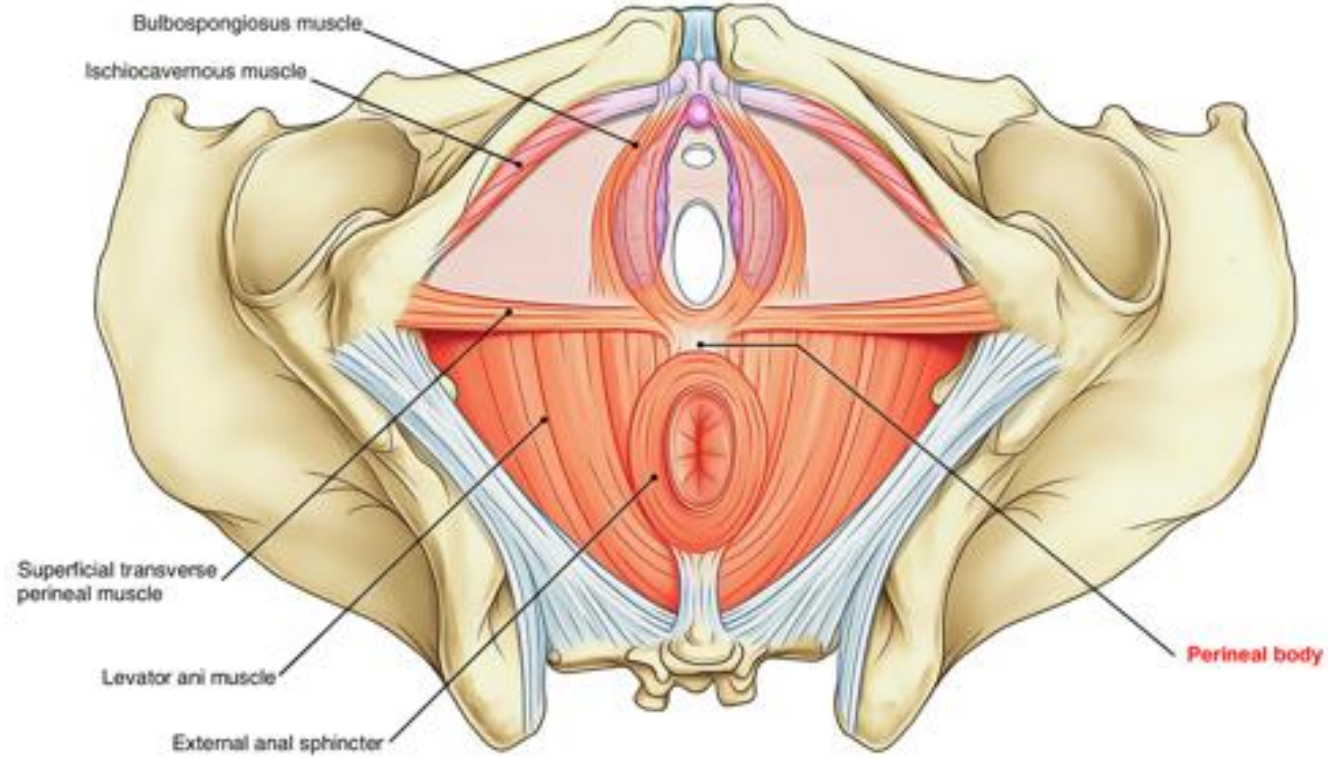


❖is a broad sheet of muscle. It is composed of **three** separate paired muscles;

1. *puborectalis,*
2. *Pubococcygeus and*
3. *iliococcygeus.*

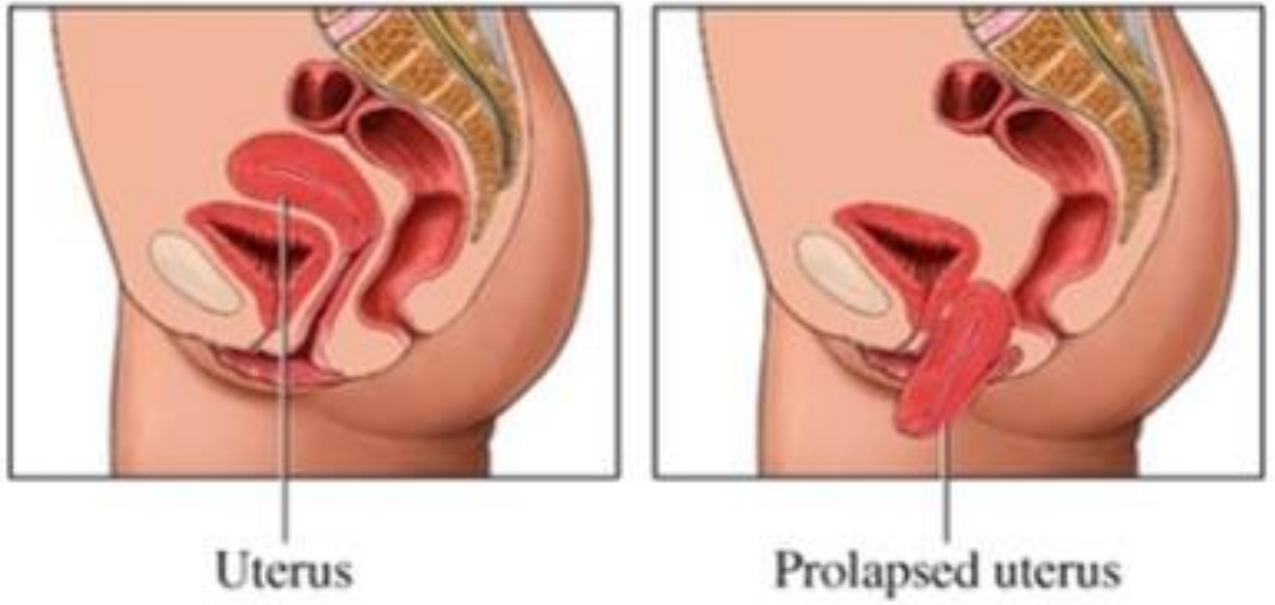
Innervated by the anterior ramus of S4 and branches of the pudendal nerve (roots S2, S3 and S4).

Perineal body



- Is a pyramidal fibromuscular mass situated in the middle of the junction of urogenital triangle and anal triangle its important for attachment of :
- *1- external anal sphenictor. 2- bulbospongiosus muscle 3- Superficial transverse perineal muscle 4 - levator ani muscle (ant part)*

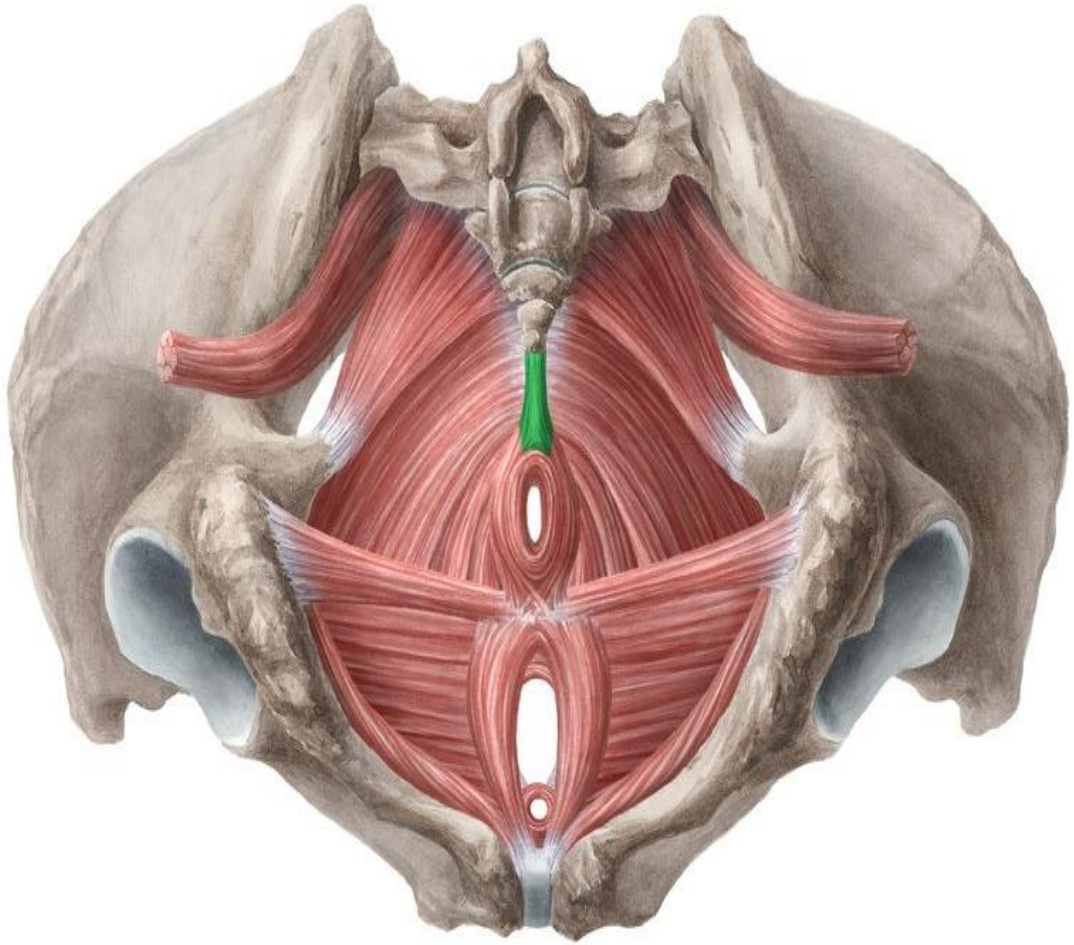
Injury of pelvic diaphragm



- The pelvic diaphragm may be injured (tearing of perineal body) during difficult childbirth.
- As a result, it becomes weak and can no longer provide sufficient support to the pelvic viscera.
- This may lead to uterine prolapse (and rectal prolapse).

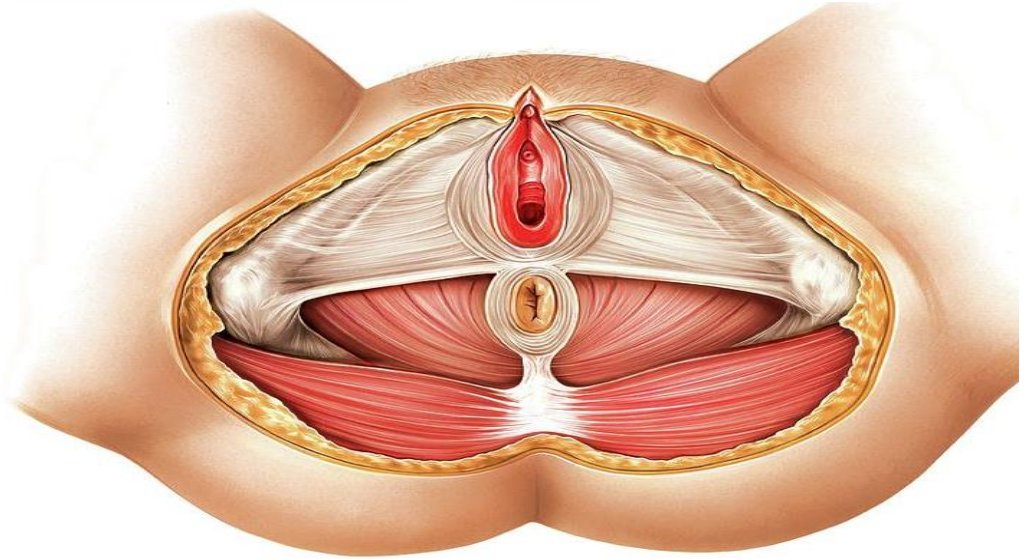
Anococcygeal Body

- The anococcygeal body is a complex musculotendinous structure
- Situated between the **anterior aspect of the coccyx** and the **posterior wall of the anorectal canal**
- Receives insertion of fibers of levator ani muscle

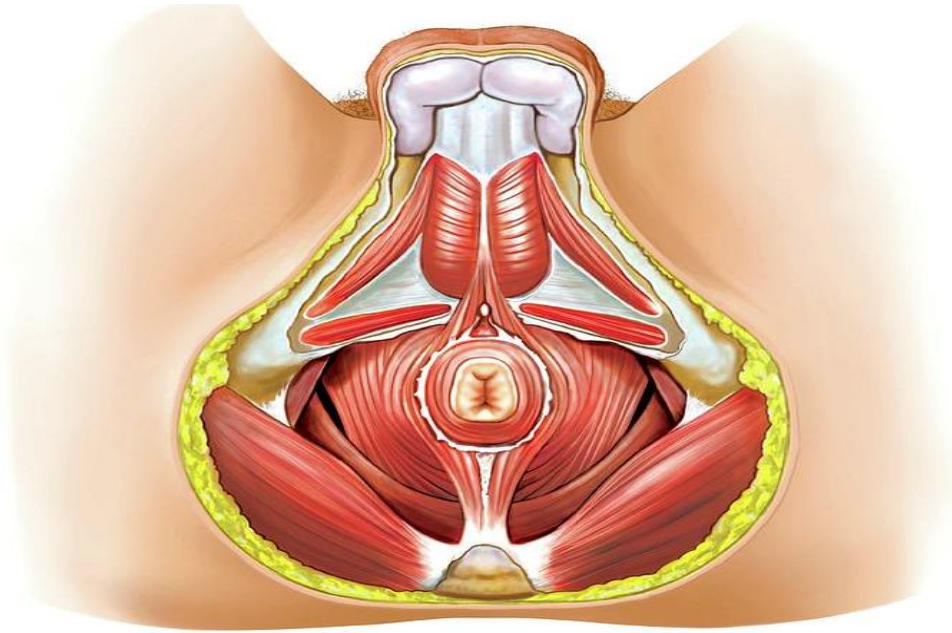


Perineum

- Perineum is the **region of the body below the pelvic diaphragm**
- It is a diamond shaped area **between the thighs**



- Contents:
  - Lower ends of urethra, vagina & anal canal
  - External genitalia
  - Perineal body & Anococcygeal body



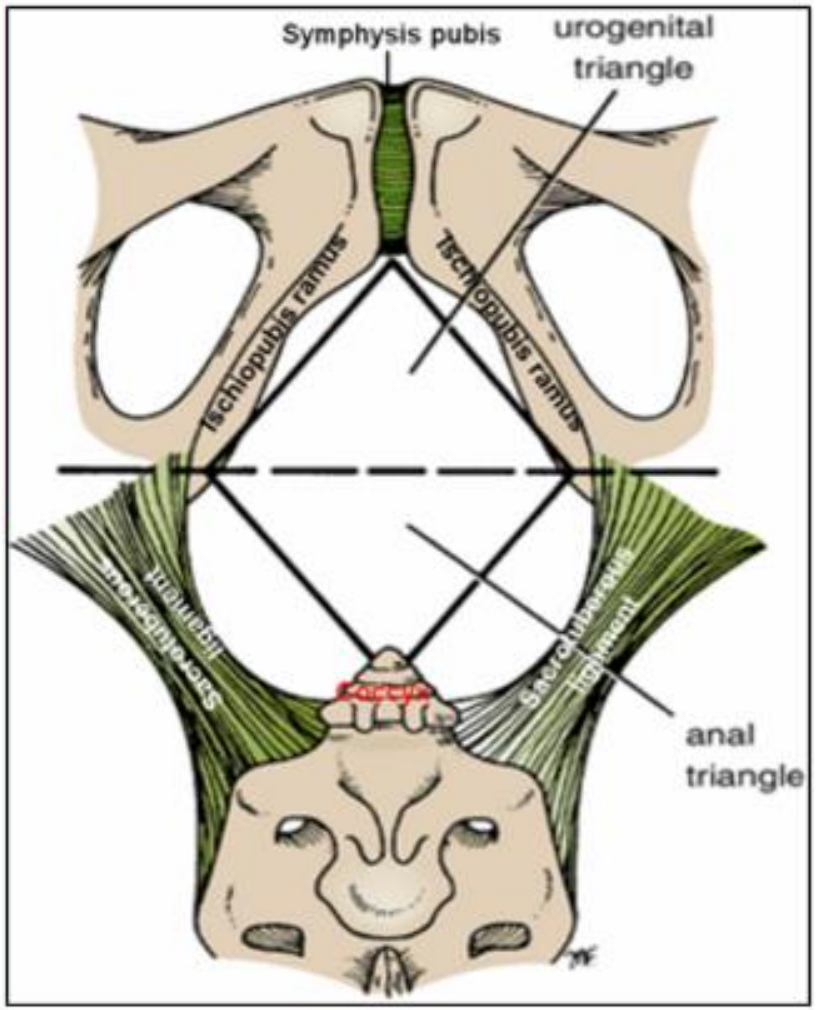
Boundaries & Division

Boundaries:

- Its bony boundaries are:
  - Anterior: Symphysis pubis.
  - Posterior: Coccyx.
  - Lateral: Ischiopubic rami, ischial tuberosities & sacrotuberous ligament.

Division:

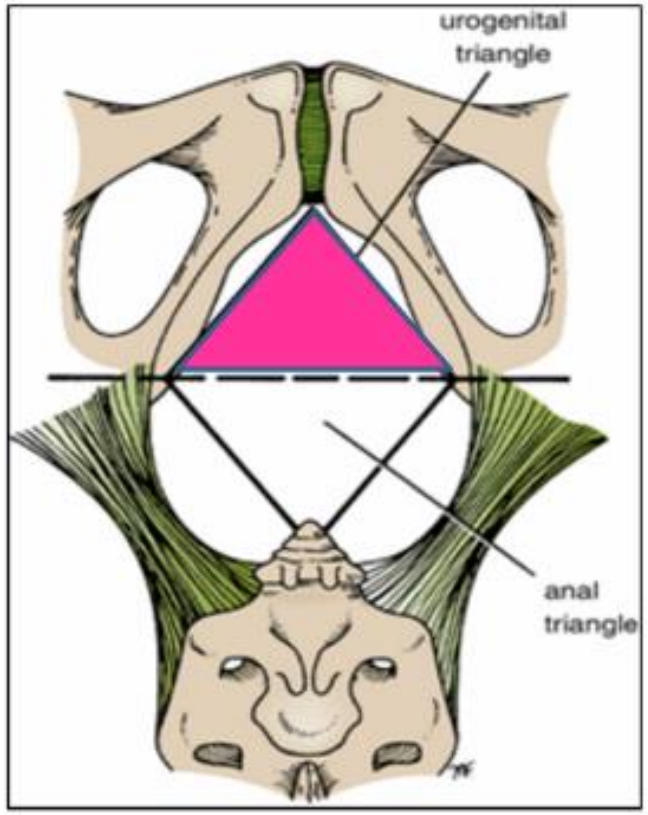
- By an imaginary line passing through two ischial tuberosities, it is divided into:
  - **Urogenital triangle** anteriorly.
  - **Anal triangle** posteriorly.

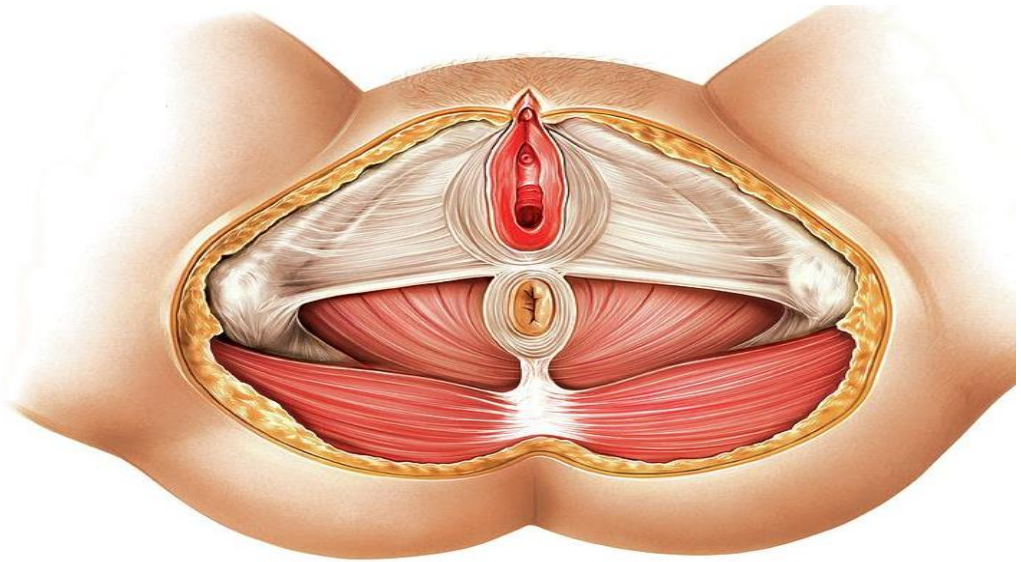


Urogenital Triangle

Boundaries :

- Anteriorly : Symphysis pubis
- Posteriorly : Transverse line passing through the 2 ischial tuberosities.
- Laterally : Ischiopubic rami & ischial tuberosities.



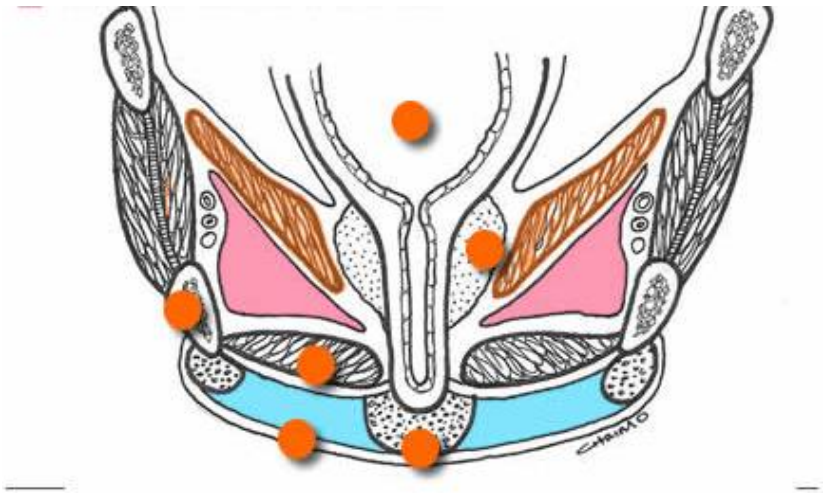


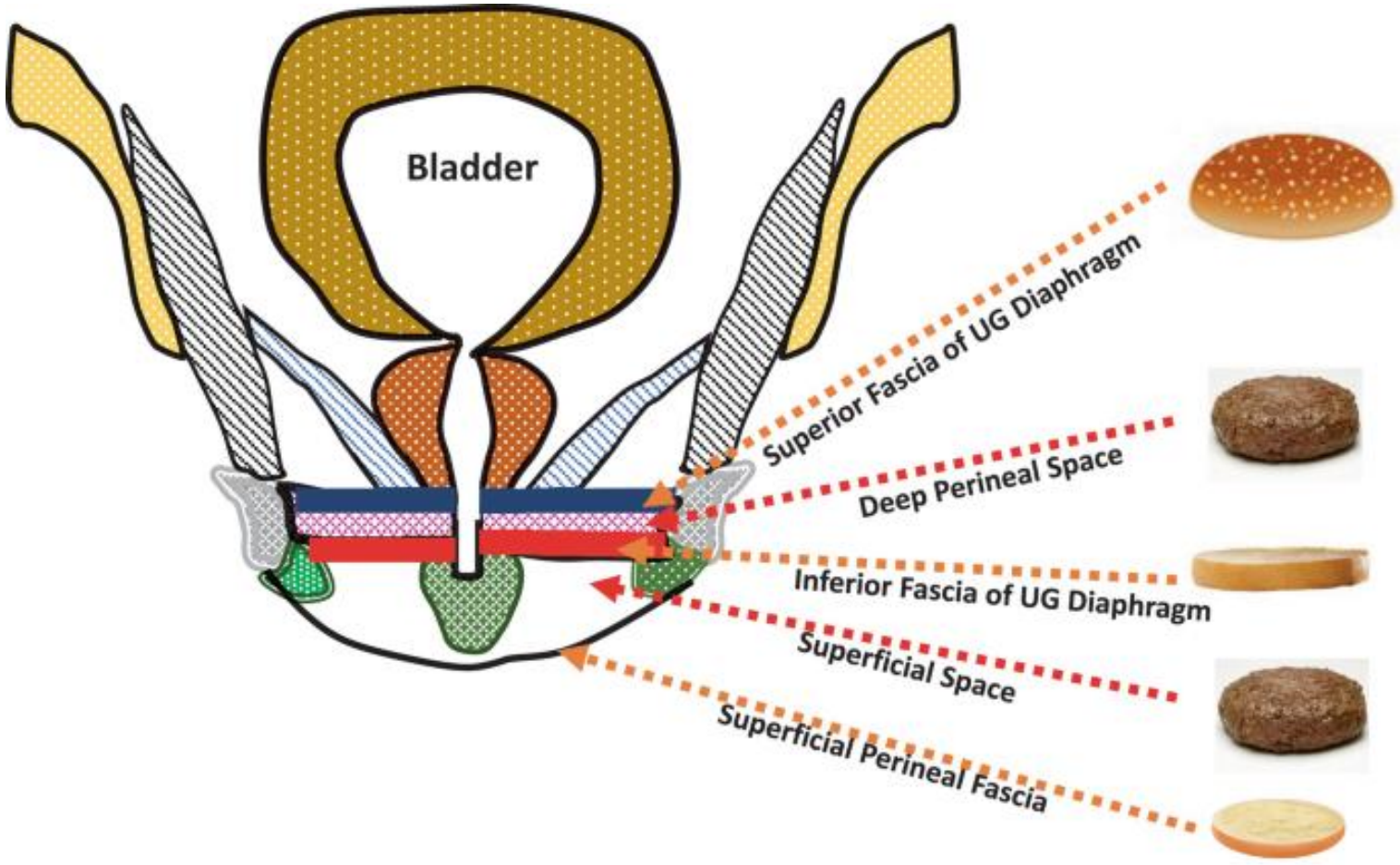
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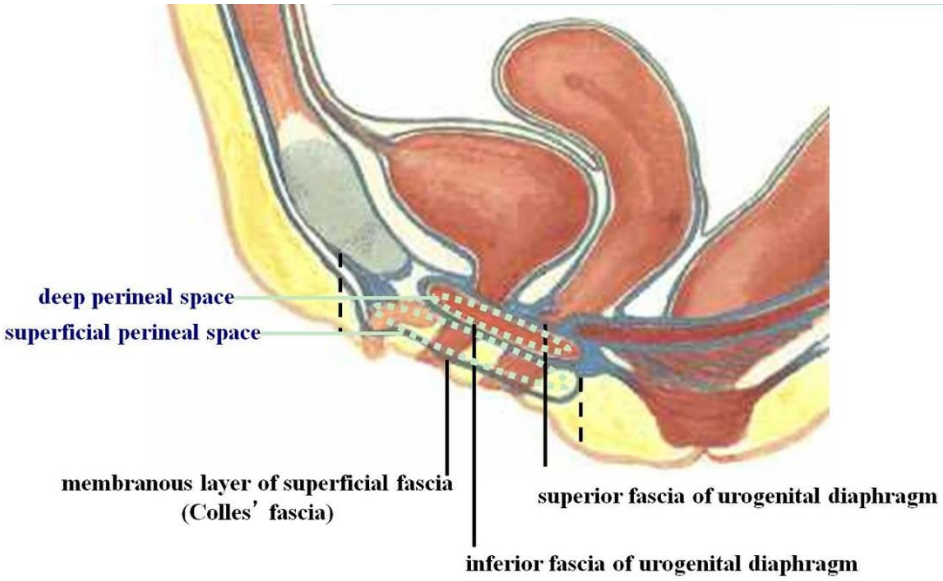
- Lower part of urethra & vagina.
- External genitalia (vulva).

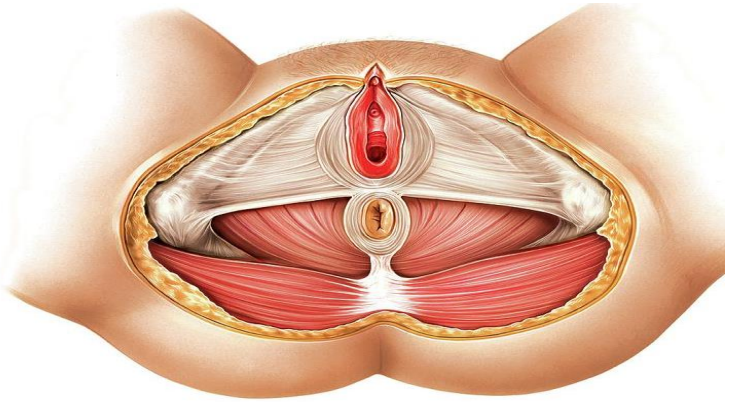
Urogenital Diaphragm

- A triangular musculofascial diaphragm located in the anterior part of the perineum
- Fills in the gap between the pubic arch
- Composed of: Sphincter urethrae and the deep transverse perineal muscles enclosed within the superior and inferior layers of fascia of the urogenital diaphragm
- The inferior layer of the fascia is formed by the perineal membrane





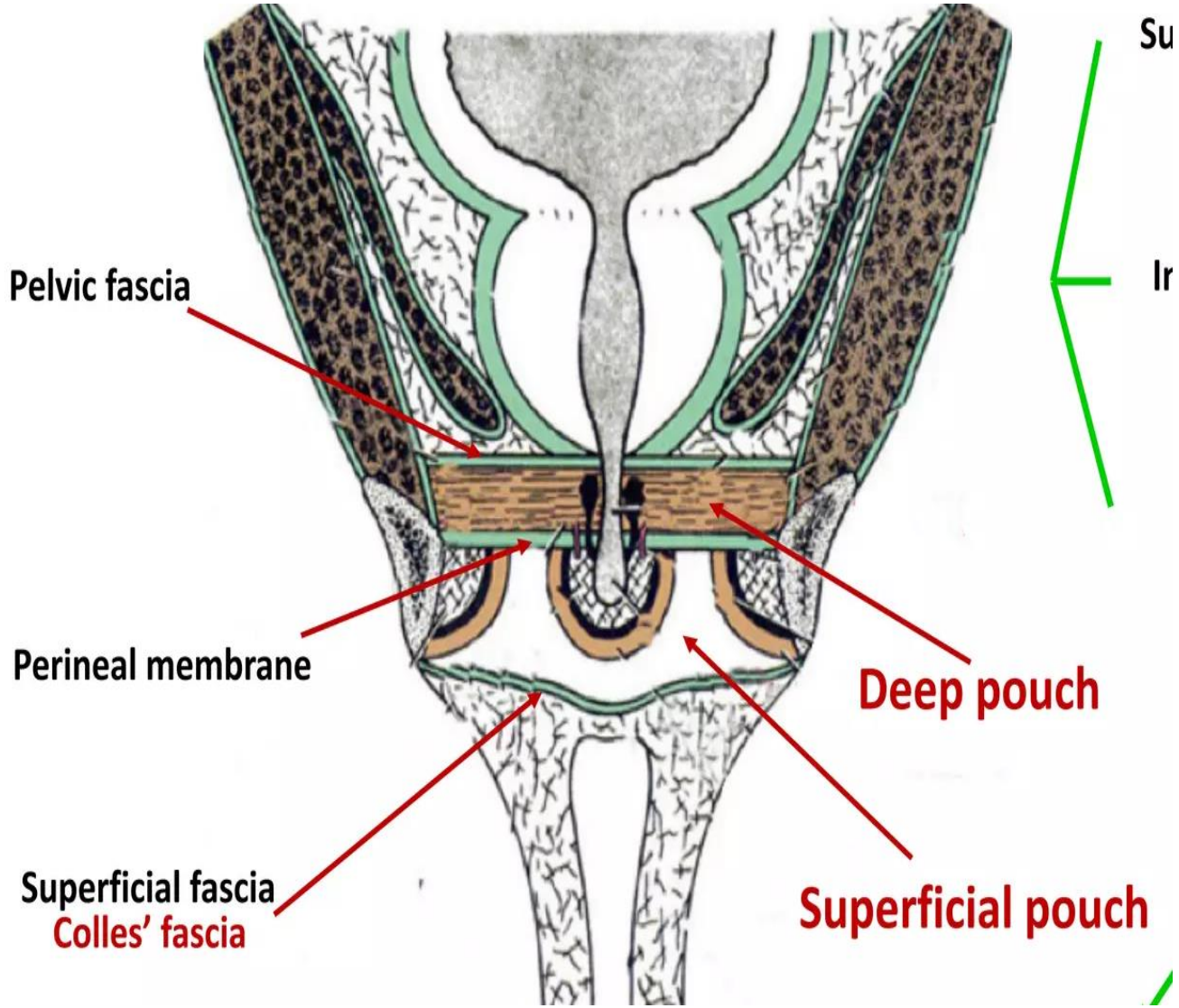




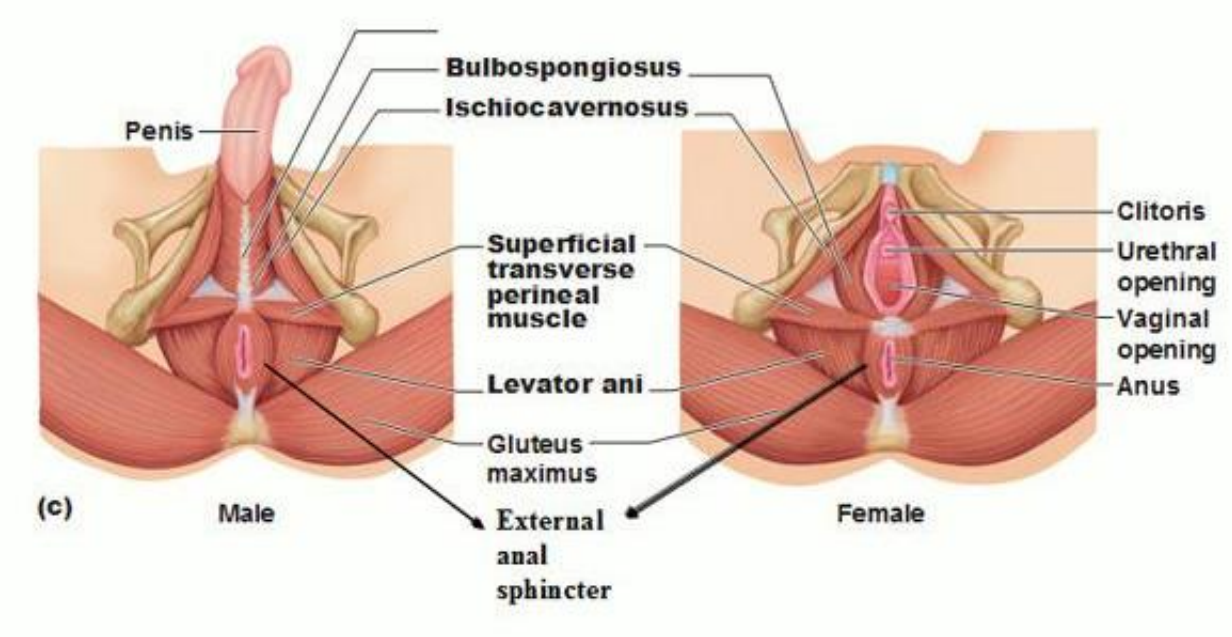
- **The urogenital triangle contains two important fascial spaces:**
  - Superficial perineal pouch.
  - Deep perineal pouch.
- The perineal membrane (Inferior fascia of UG)separates these two compartments.

Superficial Perineal Pouch

- It is the space between the deep membranous layer of superficial fascia and the perineal membrane.
- ***BOUNDARIES:***
  - *Inferiorly:* membranous layer of superficial fascia.
  - *Superiorly:* perineal membrane.
  - Laterally: ischiopubic rami



Contents of the Superficial Perineal Pouch



In Females

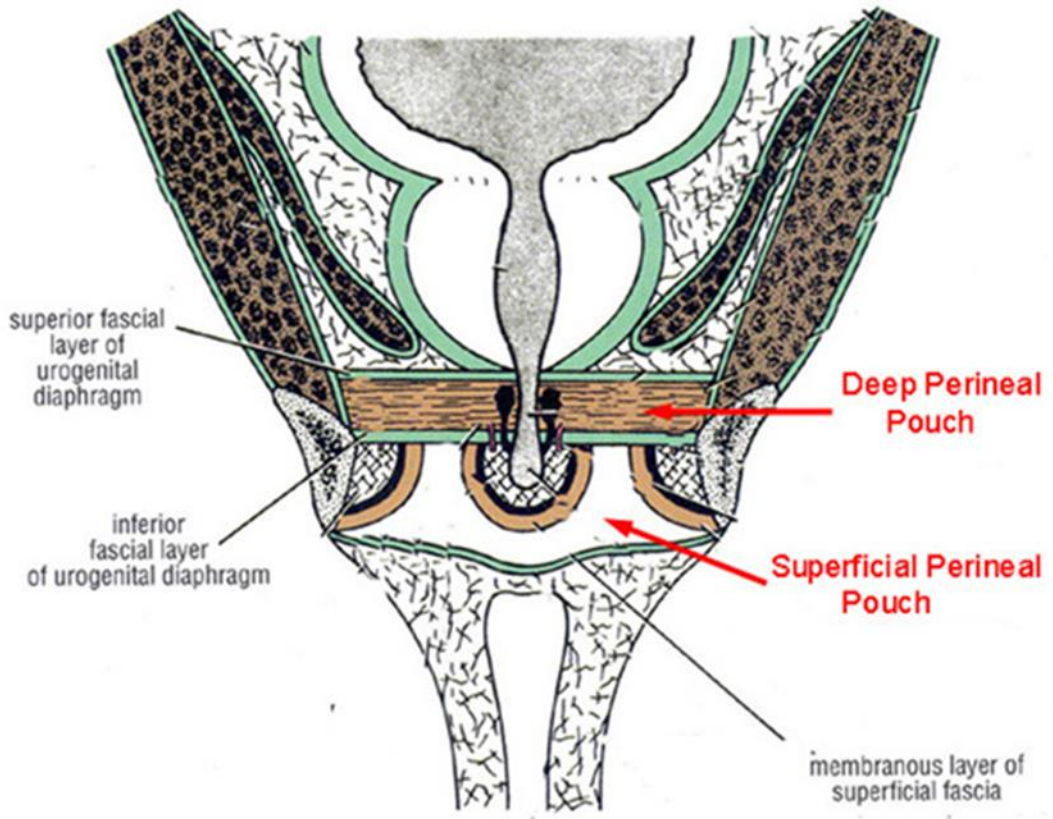
1. Erectile tissues:
    - Bulbs of the vestibule.
    - Right and left crura of the clitoris.
2. Muscles:
    - Bulbospongiosus.
    - Ischiocavernosus.
    - Superficial transverse perineal muscles.
3. Greater vestibular (Bartholin) glands.
4. Perineal branches of the internal pudendal vessels.
5. Perineal branches of the pudendal nerve.

In Males

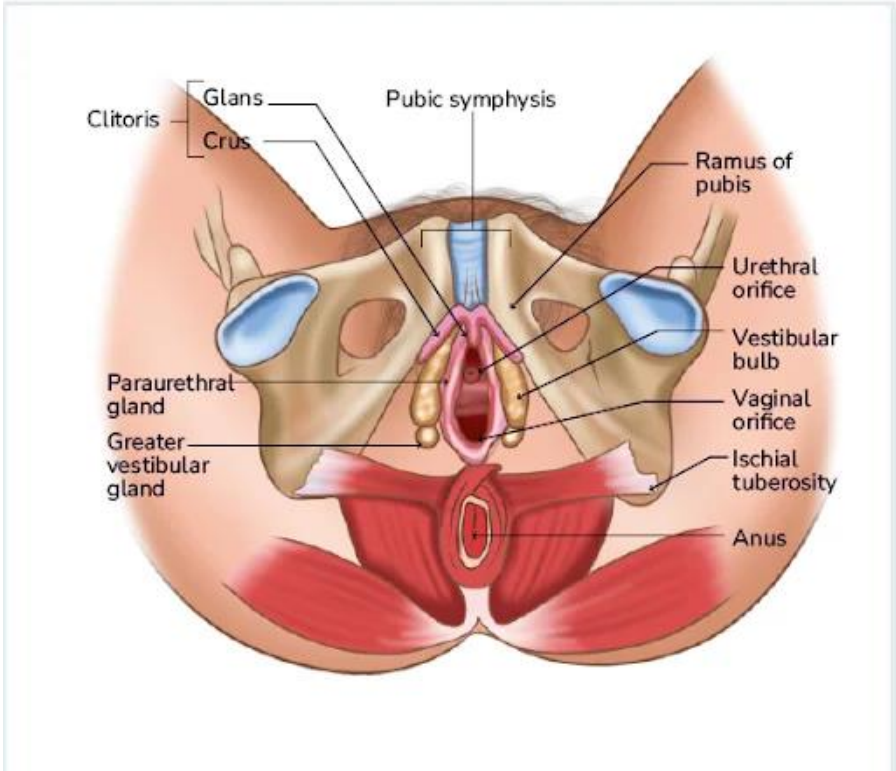
1. Root of the penis:
    - Bulb of the penis.
    - Right and left crura of the penis.
2. Muscles:
    - Bulbospongiosus.
    - Ischiocavernosus.
    - Superficial transverse perineal muscles.
3. Proximal part of the spongy (bulbar) urethra.
4. Perineal branches of the internal pudendal vessels.
5. Perineal branches of the pudendal nerve.

Deep Perineal Pouch

- It is a completely closed space deep to the perineal membrane
- *BOUNDARIES:*
  - *Inferiorly:* Inferior fascia of the urogenital diaphragm (Perineal membrane)
  - *Superiorly:* Superior fascia of the urogenital diaphragm
  - *Laterally:* Inferior portion of obturator internus facia



section of pelvis



In Males

1. Intermediate (membranous) part of the urethra.
2. External urethral sphincter muscle.
3. Deep transverse perineal muscles (variable).
4. Bulbourethral (Cowper's) glands.
5. Branches of the internal pudendal vessels, including dorsal vessels of the penis.
6. Branches of the pudendal nerve, including the dorsal nerve of the penis.

In Females

1. Part of the urethra.
2. Part of the vagina passing through the region.
3. External urethral sphincter complex:
    - Sphincter urethrae.
    - Compressor urethrae.
    - Urethrovaginal sphincter.
4. Deep transverse perineal muscles (variable).
5. Branches of the internal pudendal vessels, including dorsal vessels of the clitoris.
6. Branches of the pudendal nerve, including the dorsal nerve of the clitoris.

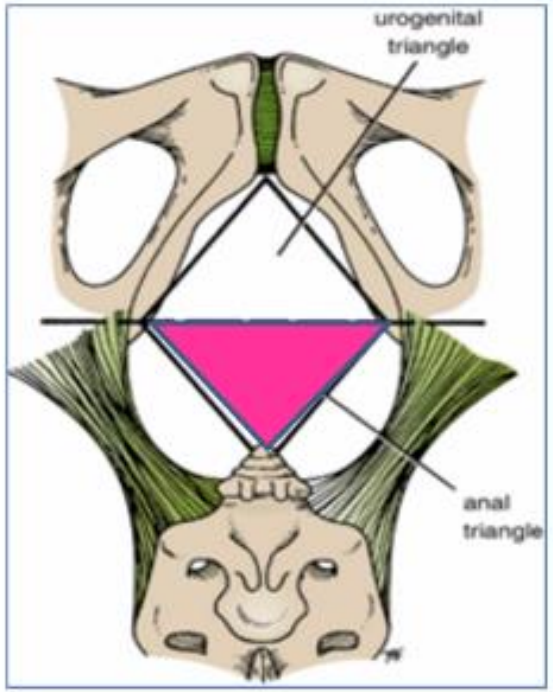
Anal Triangle

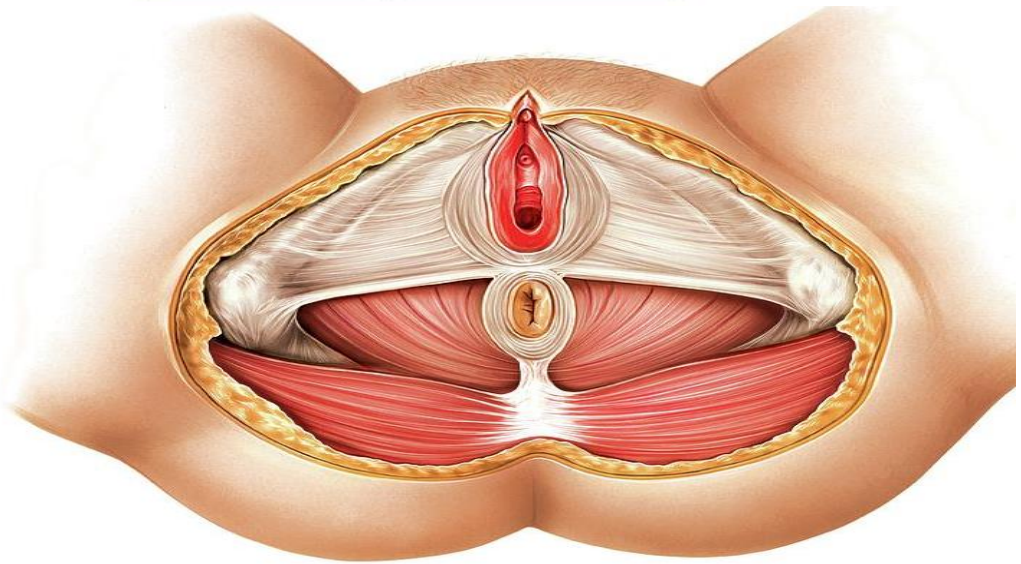
Boundaries:

- Anteriorly: Transverse line passing through the 2 ischial tuberosities.
- Posteriorly : **coccyx.**
- Laterally : **ischial tuberosity** & **sacrotuberous lig.**

Contents:

- **Lower part of Anal canal**
- **Ano-coccygeal body**
- **Ischiorectal fossa** on each side





Ischiorectal Fossa

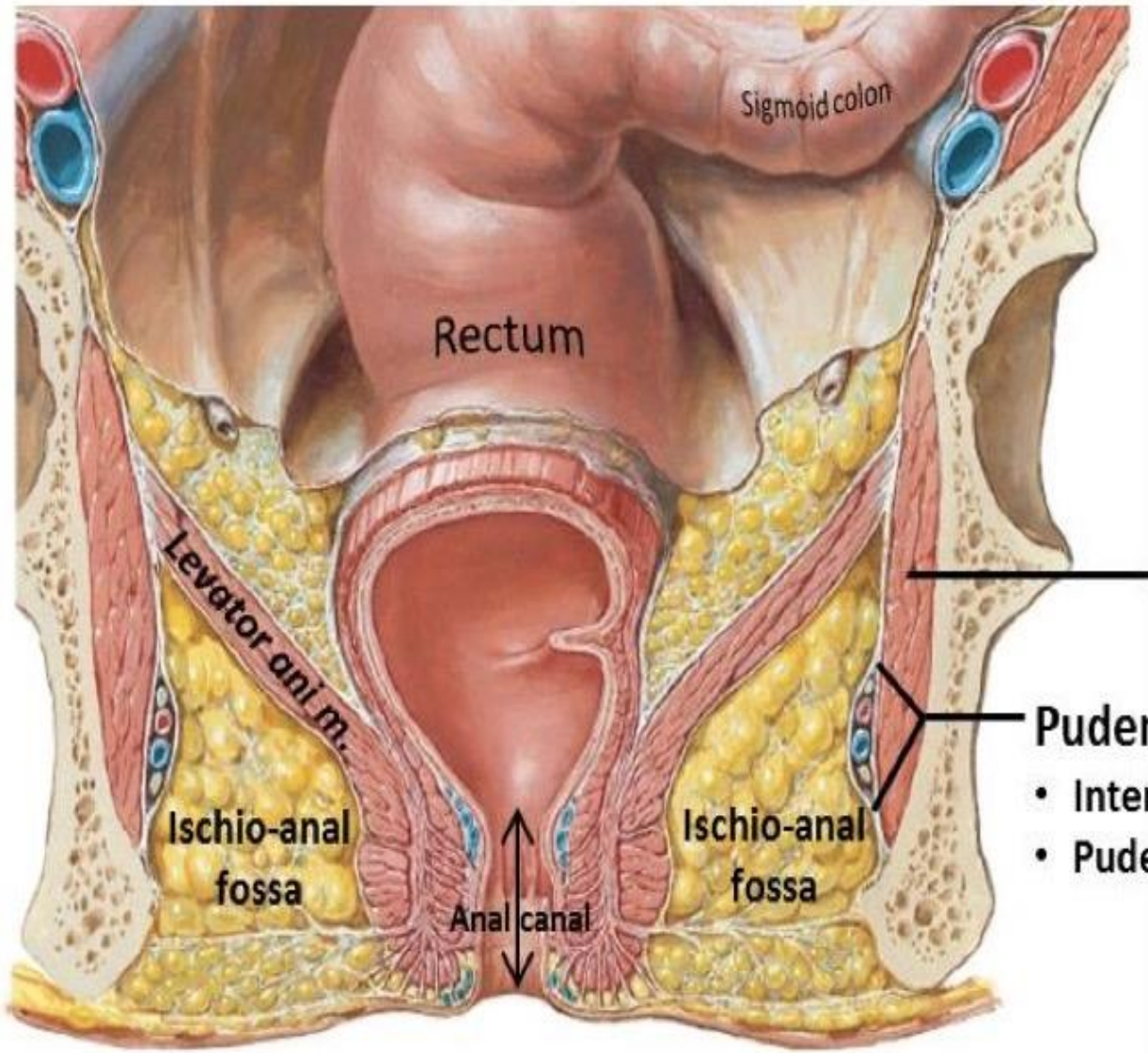
- A fascial lined wedge-shaped space on each side of the anal canal.

Boundaries:

- **Base:** Skin of the perineum.
- **Medial wall:** Levator ani & anal canal.
- **Lateral wall:** Obturator internus, covered with fascia.

Contents:

- Dense fat.
- Pudendal nerve & internal pudendal vessels within the pudendal canal
- Inferior rectal nerve & vessels crossing the fossa to reach anal canal.




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